Accelerated Math II/III Name_____ ID: 1

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HW #14 - Polynomial Inequalities

Date_____ Period____

Solve each inequality. You must show the testing of values between and beyond the zeros and the sign graph. Write your solution in interval notation.

1) $(-x+2)(2x-3) \le 0$

2) $(x-8)(x+6) \ge 0$

3) $-3x^2+17x-10 \ge 0$

4) $x^2+11x+24 \ge 0$

5) $15x^2-4x-12 > 14x^2$

6) $x^2+11x+13 < 13x+12$

7) $(-x-4)(x-2)(x-1) > 0$

8) $(x-1)^2(-2x+1) > 0$

9) $-x^3+16x^2-80x+128 \ge 0$

10) $x^3-12x^2+29x+42 > 0$

11) $(3x-5)(x+2)(-x-1)(x+6) > 0$

12) $(x-2)(x-8)^2(x+3) \le 0$

13) $-6x^4+25x^3+37x^2-64x+20 > 0$

14) $-3x^4+32x^3-59x^2-46x+48 < 0$

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15) $(x+7)(2x-1)^4 \leq 0$

16) $(x+3)(-x+3)(x+7)(x-8)(x-4) < 0$

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Answers to HW #14 - Polynomial Inequalities (ID: 1)

1) $\left(-\infty, \frac{3}{2}\right] \cup [2, \infty)$

2) $(-\infty, -6] \cup [8, \infty)$

3) $\left[\frac{2}{3}, 5\right]$

4) $(-\infty, -8] \cup [-3, \infty)$

5) $(-\infty, -2) \cup (6, \infty)$

6) Ø

7) $(-\infty, -4) \cup (1, 2)$

8) $\left(-\infty, \frac{1}{2}\right)$

9) $(-\infty, 8]$

10) $(-1, 6) \cup (7, \infty)$

11) $(-6, -2) \cup \left(-1, \frac{5}{3}\right)$

12) $[-3, 2] \cup \{8\}$

13) $\left(-2, \frac{1}{2}\right) \cup \left(\frac{2}{3}, 5\right)$

14) $(-\infty, -1) \cup \left(\frac{2}{3}, 3\right) \cup (8, \infty)$

15) $(-\infty, -7] \cup \left\{\frac{1}{2}\right\}$

16) $(-7, -3) \cup (3, 4) \cup (8, \infty)$

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